



Carleton
UNIVERSITY

**Department of
Systems and Computer Engineering**

SYSC-3310: Laboratory 7

Timer Modes

*Students are required to form a
group of 2-3 students to submit
this lab.*

*Submission is **individual based**. Each
student must submit lab deliverables. In
each individual submission, the
names/IDs of ALL students in the group
must be included.*

Lab 7 – Timer Modes

In this lab, you will exercise 2 things: programming interrupts and configuring/using the timers. It's assumed you are already well versed in I/O. The goal is to refresh and practice Timer configuration and ISR usage.

Things to look out for:

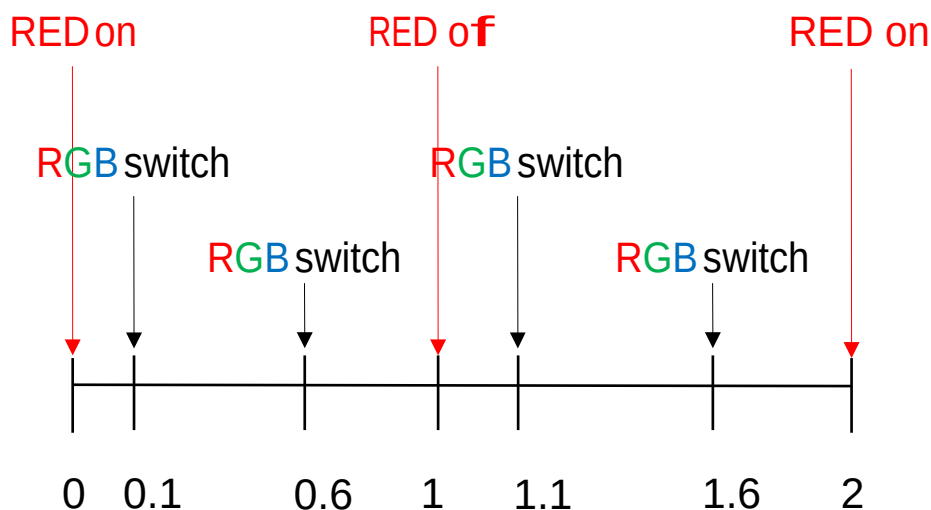
- Configuring timer(s) correctly.
- Configuring interrupts correctly.
- Implementing ISRs correctly.
- All other required configurations
- Correct program behavior

Create a new project in Keil, named "Lab5".

In this project, you should:

- 1 - Disable the Watchdog timer!
- 2 - Configure the LEDs (P1.0 and P2.0, P2.1, P2.2) as outputs.
- 3 - Initialize LEDs states (all turned off).
- 4 – Configure 2 Timers A (16 bits) accordingly.
- 5 – Implement switch and timer ISRs

We want one Red LED to blink (on/off) every second, and the second LED (RGB) to blink (change to next color) every 0.5 seconds. The two LEDs should be out of phase by precisely 0.1 seconds (see diagram below). Use TimerA0 in UP mode (and corresponding interrupts) to trigger the RED LED and TimerA1 in UP/DOWN mode (and corresponding interrupts) to trigger the RGB LED.



Deliverables

To get the marks for this lab, submit the following:

- 1) Your project folder in a single zip file. Name your project zip file like this:
Lab7Proj_<your_ID>_<LastName>.zip
Your zip folder MUST include the entire Keil project folder
- 2) A video that shows a DEMO of your project testing on the board.
- 3) A one-page abstract of what you have done and learned in this lab. Write ALL group students' names and ID on this abstract page.

Due Date

The lab deliverable is **one-weeks** due from your lab session date.